



HR Analytics: Employee Attrition Analysis

Using [MySQL](#) and [Power BI](#) to analyze employee attrition and workforce trends.

DATA ANALYST INTERNSHIP PROGRAM

UNIFIED MENTOR

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Project Overview

The **HR Analytics Dashboard** project bridges the gap between raw employee data and strategic decision-making. By combining MySQL for data management and Power BI for visualization, this project turns workforce records into a clear, interactive story about attrition, department performance, and salary trends.

Analyze Attrition

Identify patterns in employee turnover and understand which factors drive attrition across the organization.

Workforce Trends

Study how department, job role, overtime, and salary influence employee behavior and retention.

Business Insights

Transform raw HR data into meaningful metrics that support data-driven HR strategy and planning.

Interactive Dashboard

Build a dynamic Power BI dashboard that allows stakeholders to explore data visually and in real time.



Project Objectives

This project was designed with five clear goals — each one building toward a complete, end-to-end HR analytics solution that delivers real value to HR teams and leadership.

1

Analyze Employee Data

Use MySQL to query and clean the HR dataset, then connect it to Power BI for structured analysis.

2

Calculate Attrition Metrics

Derive key KPIs including **Attrition Rate** and **Attrition Count** using DAX formulas in Power BI.

3

Study Overtime & Department Impact

Examine how overtime hours and department assignments correlate with employee turnover rates.

4

Create Interactive Visualizations

Design charts, cards, and filters in Power BI that make data exploration intuitive for non-technical users.

5

Support HR Decision-Making

Deliver a polished dashboard that HR managers can use to identify risk areas and plan retention strategies.

Tools & Technologies

A focused, industry-standard toolkit was used to build this project — from database creation to final dashboard delivery. Each tool plays a distinct role in the analytics pipeline.



MySQL (XAMPP)

Used to create and manage the HR database locally. XAMPP provides a reliable development environment for running MySQL queries and importing datasets.



DAX

Data Analysis Expressions (DAX) is the formula language used in Power BI to create custom calculations such as Attrition Rate, Total Headcount, and department-level metrics.



Power BI Desktop

The primary visualization platform. Power BI connects to MySQL, imports data, and enables the creation of interactive reports and dashboards with rich visual elements.



phpMyAdmin

A web-based interface for managing MySQL databases. Used to import the CSV dataset, inspect tables, and verify data integrity before connecting to Power BI.

Dataset Description

Green Destinations HR Dataset

A comprehensive employee records dataset covering demographic, job, and compensation information for a mid-size organization.

1,470

Total Employees

Key Data Fields

The dataset includes the following attributes, each enabling a different dimension of HR analysis:

- **Age & Gender** – Demographic segmentation for diversity and attrition analysis
- **Department & Job Role** – Organizational structure and role-level breakdowns
- **Salary** – Compensation data for pay equity and role-based comparisons
- **Overtime** – Binary flag indicating whether an employee works overtime
- **Attrition** – Target variable indicating whether an employee has left the company

Database Creation & Setup

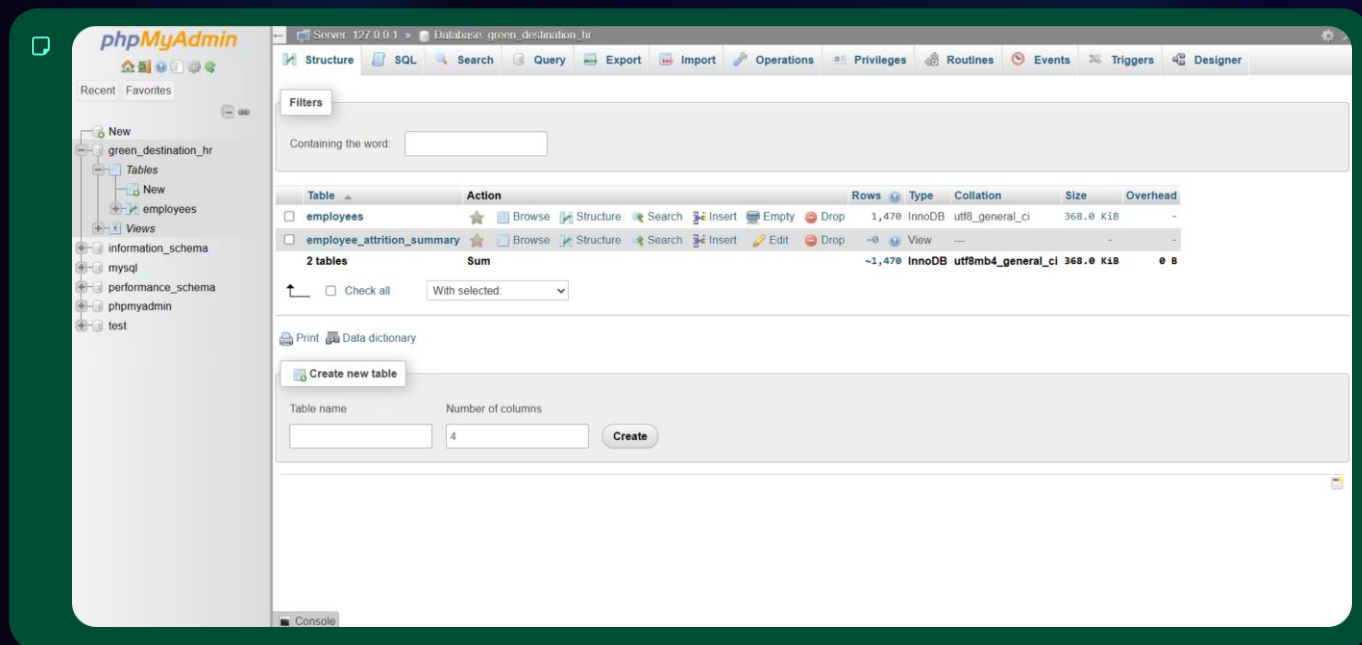
The foundation of this project is a well-structured MySQL database. Using [XAMPP](#) and [phpMyAdmin](#), the HR dataset was imported, cleaned, and prepared for seamless integration with Power BI.

Database Setup

A dedicated **HR database** was created in MySQL via XAMPP. Tables were structured to match the dataset schema, ensuring clean column names and appropriate data types for each field.

Data Import via phpMyAdmin

The Green Destinations CSV was imported through phpMyAdmin's intuitive interface. Data integrity checks were performed to confirm all 1,470 employee records loaded correctly with no missing values.



Power BI Dashboard

The final deliverable is a fully interactive [Power BI dashboard](#) that surfaces the most critical HR metrics at a glance. Each visual is designed to answer a specific business question, making it easy for HR managers to spot trends and take action.

PowerBI Connection

Power BI Desktop imported the MySQL data into its internal model using Import Mode. Since the native connector encountered compatibility issues, the dataset was extracted from MySQL and loaded directly into Power BI, allowing for optimized performance and reliable dashboard functionality.



Total Employees

Headcount card showing the full workforce size of 1,470 employees at a glance.



Attrition Count

Total number of employees who have left the organization, calculated using DAX.



Attrition Rate

Percentage-based KPI showing overall turnover rate across the entire workforce.



Department Analysis

Breakdown of attrition and headcount by department to identify high-risk areas.



Overtime Analysis

Comparison of attrition rates between employees who work overtime and those who don't.



Salary Analysis

Salary distribution by job role and department, highlighting compensation patterns.



Key Insights



Overtime is a major attrition driver —

Employees working overtime showed significantly higher turnover rates, suggesting workload management is critical for retention.



Department-level variation —

Attrition was not uniform across departments. Certain teams experienced disproportionately higher turnover, pointing to localized cultural or operational issues.



Salary gaps by role —

Compensation varied meaningfully across job roles, providing a baseline for pay equity analysis and benchmarking.

Conclusion

This project successfully delivered an **end-to-end HR Analytics Dashboard** — from database creation in MySQL to a polished, interactive Power BI report. It demonstrates how structured data analysis can directly support HR strategy.

Three key achievements:

SQL & Power BI Skills

Strengthened proficiency in querying, data modeling, DAX, and dashboard design.

Business Impact

Generated actionable insights that HR teams can use to reduce attrition and improve workforce planning.

Portfolio-Ready Project

A complete, documented analytics project showcasing real-world data analysis capabilities.